MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC TEST – 06 (POLYNOMIALS)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 2: Polynomials** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in Section D.
2. **Section A** comprises 5 questions (4 MCQs and 1 Assertion-Reason) of 1 mark each.
3. **Section B** comprises 3 VSAQ questions of 2 marks each.
4. **Section C** comprises 3 SA questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS questions of 5 marks each.

SECTION A: Objective & MCQs (5 × 1 = 5 Marks)

Q1. If the graph of quadratic polynomial $ax^2 + bx + c$ cuts the negative direction of the y-axis, then the sign of c is: [1]

(A) positive (B) negative (C) zero (D) depends on a

Q2. If the graph of $y = ax^2 + bx + c$ opens upwards, then: [1]

(A) $a > 0$ (B) $a < 0$ (C) $a = 0$ (D) $c > 0$

Q3. The zeroes of the quadratic polynomial $6x^2 - 3 - 7x$ are: [1]

(A) $\frac{3}{2}, -\frac{1}{3}$ (B) $-\frac{3}{2}, \frac{1}{3}$ (C) $\frac{3}{2}, \frac{1}{3}$ (D) $-\frac{3}{2}, -\frac{1}{3}$

Q4. If zeroes of the quadratic polynomial $ax^2 + bx + c$ are equal, then: [1]

(A) c and a have opposite signs (B) c and b have opposite signs
(C) c and a have the same sign (D) c and b have the same sign

Q5. **Assertion (A):** If α, β are zeroes of $x^2 - 6x + k$ and $3\alpha + 2\beta = 20$, then $k = -16$.
Reason (R): For $x^2 - 6x + k$, $\alpha + \beta = 6$ and $\alpha\beta = k$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Questions (3 × 2 = 6 Marks)

Q6. If the product of zeroes of $f(x) = x^2 - 4x + k$ is 3, find the value of k. [2]

Q7. If α, β are zeroes of $f(x) = x^2 - p(x + 1) - c$, prove that $(\alpha + 1)(\beta + 1) = 1 - c$. [2]

Q8. If one zero of $f(x) = 3x^2 + 8x + k$ is reciprocal of the other, find the value of k. [2]

SECTION C: Short Answer Questions (3 × 3 = 9 Marks)

Q9. If α and β are zeroes of $f(x) = 2x^2 + 11x + 5$, find the values of: [3]

(i) $\alpha^2 + \beta^2$ (ii) $\frac{1}{\alpha} + \frac{1}{\beta}$ (iii) $\alpha^3 + \beta^3$

Q10. If α, β are the zeroes of the quadratic polynomial $ax^2 + bx + c$, express $\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}$ in terms of a, b, c. [3]

Q11. If α, β are zeroes of $f(x) = 2x^2 - 5x + 7$, find a quadratic polynomial whose zeroes are $2\alpha + 3\beta$ and $3\alpha + 2\beta$. [3]

SECTION D: Long Answer / HOTS Questions (2 × 5 = 10 Marks)

Q12. If α, β are the zeroes of $f(x) = x^2 - 4x + 1$, evaluate: [5]

(i) $\alpha^2 + \beta^2$ (ii) $\alpha^3 + \beta^3$ (iii) $\alpha^4 + \beta^4$ (iv) $\frac{\alpha}{\beta^2} + \frac{\beta}{\alpha^2}$

Q13. If the quadratic polynomial $f(x) = ax^2 + bx + c$ satisfies $f(1) = 0$, $f(2) = 3$, and $f(3) = 8$, determine the coefficients a, b, c. Hence find the zeroes of the polynomial. [5]

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